

EXPERIMENT NO. : 2DATE: 13/08/2026

1. AIM

Demonstration of Processing Steps Involved in the cleaning of Glass/Silicon Wafers

2. TOOL

Glass/Silicon wafer or substrate

Ultrasonic sonicator

Beakers or sample holders

DI water

Acetone

Isopropyl alcohol (IPA)

Nitrogen blow-off or air-drying facility

PPE such as gloves and goggles

3. THEORY

Cleaning is a critical step in semiconductor fabrication because the wafer surface must be completely free from any contamination before processes such as lithography or deposition can be performed. Even tiny particles or dust can negatively impact the final device performance.

Different cleaning methods use a predefined sequence of chemicals, each selected for removing a specific type of impurity like organic residues, salts, dust, or other unwanted materials. The order and timing of these chemicals form the cleaning recipe.

During this demonstration, a glass substrate was cleaned using various chemicals along with an ultrasonic sonicator. The ultrasonic vibrations help dislodge fine particles from the surface, while the chemicals dissolve or wash away remaining contaminants. This process ensures that the substrate becomes properly cleaned, smooth, and suitable for the next fabrication steps.

4. PROCEDURE

1. Prepare the Sonicator & Samples

- Fill the sonicator bath with water (or the appropriate solvent if needed).
- Place samples in a beaker or holder (ensure they are sonicator-safe).

2. First Cleaning Cycle (DI Water)

- Submerge samples in a beaker filled with DI water.
- Run the sonicator for 15 minutes at recommended power.
- Removes loose particles and water-soluble contaminants.

3. Rinse & Inspect

- Remove samples and rinse with fresh DI water.
- Check for visible residues; repeat if needed.

4. Second Cleaning Cycle (Acetone)

- Place samples in a clean beaker with acetone (for organic residues).
- Sonicate for 15 minutes (acetone evaporates quickly).

5. Rinse & Transition to IPA

- Discard acetone, rinse samples with fresh IPA to avoid water streaks.

6. Final Cleaning Cycle (IPA)

- Sonicate samples in IPA for 15 minutes for degreasing & drying.
- Ensures no solvent residues remain.

7. Dry & Verify Cleanliness

- Remove samples, air-dry or use nitrogen blow-off.
- Inspect under a microscope if needed.

Safety Notes:

- Wear PPE (gloves, goggles).
- Work in a fume hood for acetone/IPA.
- Dispose of solvents properly.

6. OBSERVATION

The glass substrate was cleaned sequentially using DI water, acetone, and IPA with ultrasonic sonication. Loose particles, water-soluble contaminants, and organic residues were removed, and the substrate appeared clean and free from visible residues after drying.

7. CONCLUSION

The cleaning process for the glass/silicon substrate was successfully demonstrated using DI water, acetone, IPA, and ultrasonic sonication. Each cleaning stage removed specific types of contaminants and prepared the surface for further semiconductor processing. The experiment demonstrated the importance of proper wafer cleaning in achieving a clean and suitable substrate.